Save a copy

Lesson 1: Database essentials

KS4 - Databases and SQL

National Centre for Computing Education

How is data stored?

IN the UK you can search for vehicle information on the DVLA website.

How is data stored?

Introduction

For this activity you will be searching for information about UK registered vehicles using the DVLA website.

Task Search for a vehicle

Step 1

Visit the DVLA website (ncce.io/ks4-DVLA) and choose **Start Now**.

Step 2

Enter the **license plate number** below, choose **yes** and click **continue**.

D504 YNV

Step 3

Fill in the details below using the information displayed about the vehicle:

Vehicle make:	
Year of manufacture:	
Is the vehicle taxed?	

How is data stored?

Question

How do you think the vehicle data has been stored?

In a table?

In a database?

In a structure that is easy to search?

✓ Taxed
Tax due:
1 April 2021

✓ MOT
Expires:
22 March 2021

▶ Incorrect tax status?

▶ Incorrect MOT status?

If you've just bought this vehicle the tax or SORN doesn't come with it. You'll need to tax it before driving it.

Vehicle Details

Vehicle make	**VOLKSWAGEN**
Date of first registration	**March 2009**
Year of manufacture	**2009**
Cylinder capacity	**1896 cc**
CO_2 emissions	**0 g/km**
Fuel type	**DIESEL**
Euro status	**Not available**
Real Driving Emissions (RDE)	**Not available**

DVLA services

- Tax your vehicle
- Register your vehicle as off the road (SORN)
- Change your vehicle's tax class
- Tell DVLA you've sold, transferred or bought a vehicle
- Report an untaxed vehicle
- Check you're not buying a stolen vehicle

How is data stored?

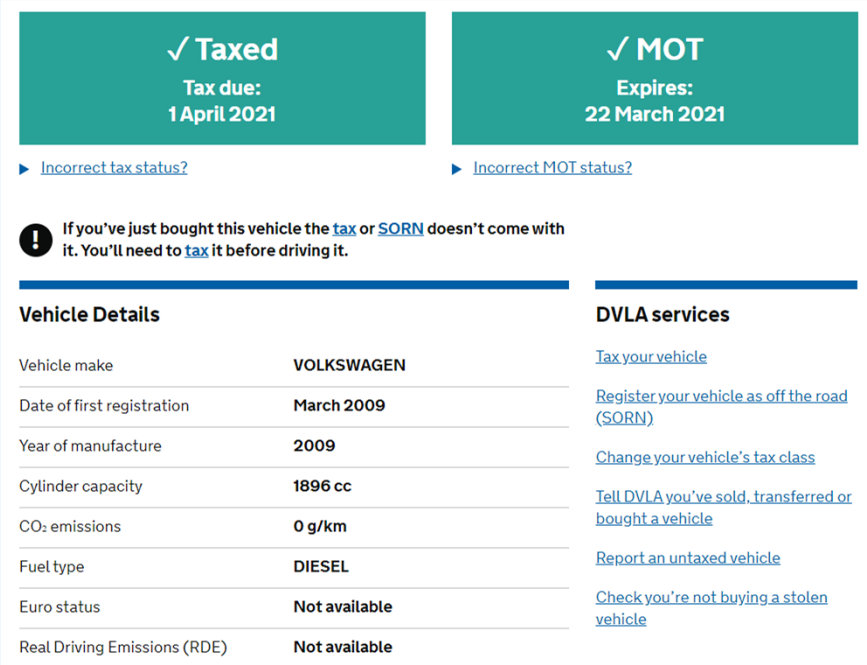
Question

How many vehicles do you think the DVLA might have in their **records**?

Millions?

Every registered car in the country?

37.9 million



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Tax your vehicle

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Change your vehicle's tax class

Tell DVLA you've sold, transferred or bought a vehicle

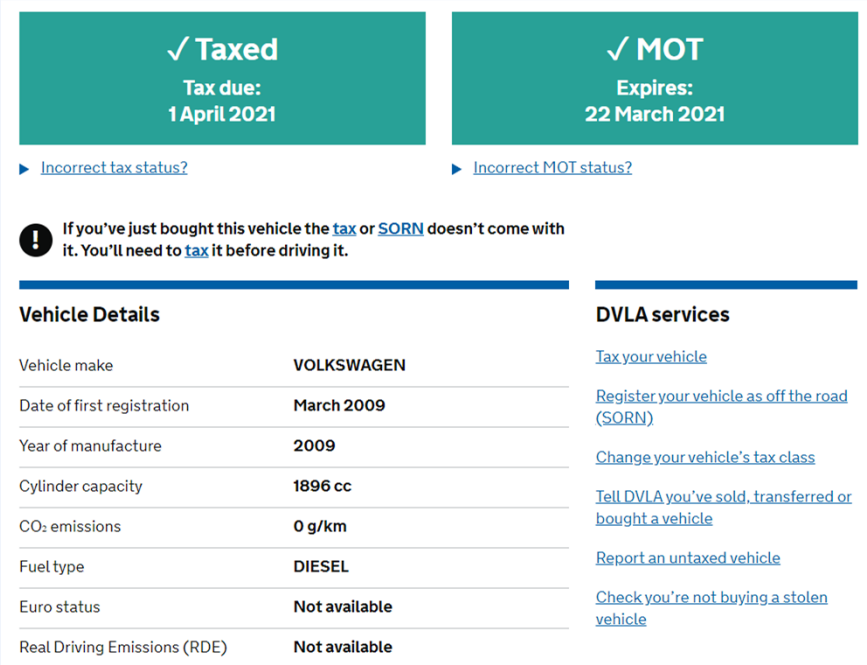
Report an untaxed vehicle

Check you're not buying a stolen vehicle

How is data stored?

When large amounts of data need to be collected and organised it is often stored in a **database**.

A **database** is a structured set of data. It is structured in **tables** that are easy to search and update.



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DVLA services

Tax your vehicle

Register your vehicle as off the road (SORN)

Change your vehicle's tax class

Tell DVLA you've sold, transferred or bought a vehicle

Report an untaxed vehicle

Check you're not buying a stolen vehicle

How is data stored?

Here you can see a single **record** from a **database**.

A **record** is a collection of data for one object, person or thing.

In this case it is a record for a vehicle.

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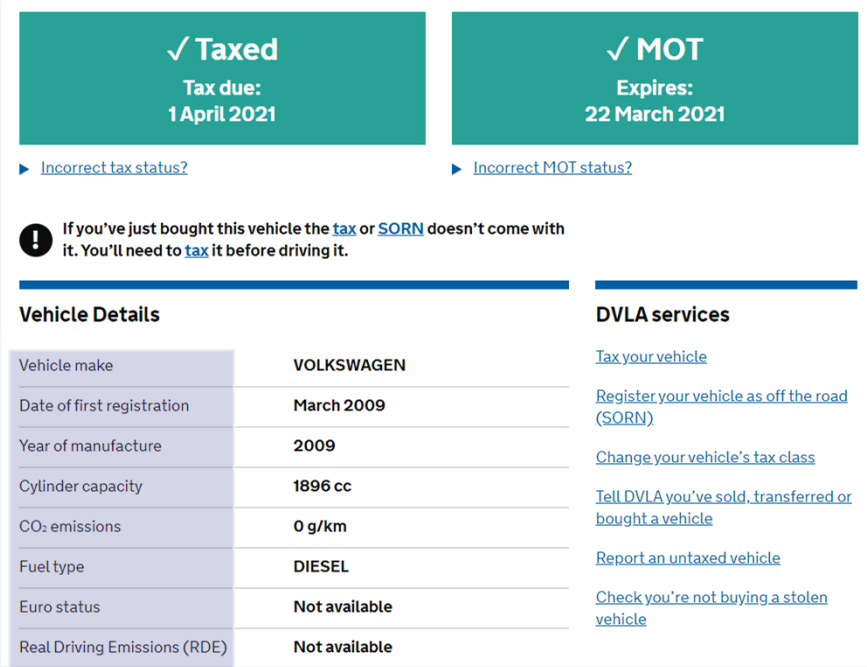
- Tax your vehicle
- Register your vehicle as off the road (SORN)
- Change your vehicle's tax class
- Tell DVLA you've sold, transferred or bought a vehicle
- Report an untaxed vehicle
- Check you're not buying a stolen vehicle

How is data stored?

Fields are used in a database to provide category headings for each piece of data in the database.



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Lesson 1: Database essentials

In this lesson, you will:

- Describe a database
- Define database key terms (table, record, field, primary key, foreign key)
- Describe a flat file database
- Describe a relational database

Database key terms

Here is a **flat file database**. A flat file database is a database that contains a **single table**.

tblTracks

TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

Database key terms

A **table** holds a collection of **records** for a particular theme.

In this case it is a collection of music tracks.

tblTracks

TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

Database key terms

The naming convention for a **table** always begins with a lowercase **tbl** followed by the name of the table.

tblTracks

TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

Database key terms

A **table** contains a collection of **records**.

Each **record** contains the data for one object, person, or place.

In this example, each record is a single song track.

tblTracks

TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

Database key terms

A **table** has a number of **fields**. These are the categories for each piece of data in the table.

This table has **four** fields.

- **TrackID**
- **Title**
- **Artist**
- **Genre**

Fields

TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

Database key terms

Each **record** in the table needs a **unique** identifier.

A **primary key** is used to give each record a **unique** code.

tblTracks

TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

Database key terms

A **primary key** can never be repeated.

This ensures that each **record** in the table is **unique**. This is very important when dealing with thousands or even millions of records.

tblTracks

TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

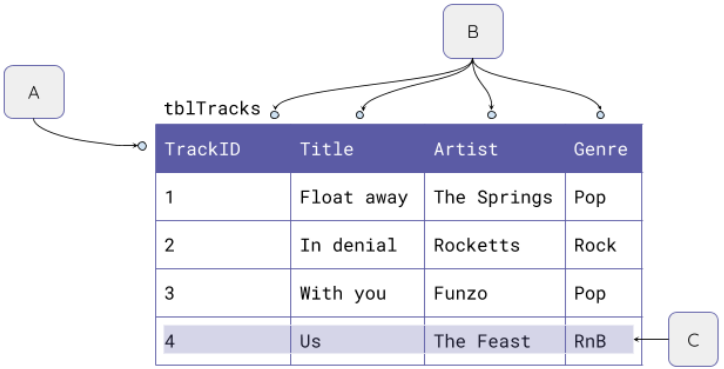
Database key terms

Complete the **A1 Worksheet** which is a quick quiz on the database key terms.

Database key terms quiz

Task Complete the quiz

A database is a structured collection of data that can easily be searched and updated. Here are some questions about database key terms.



TrackID	Title	Artist	Genre
1	Float away	The Springs	Pop
2	In denial	Rocketts	Rock
3	With you	Funzo	Pop
4	Us	The Feast	RnB

An inefficient flat file database

TrackID	Title	Artist	Genre	DloadID	Date	Time	MemberID	Firstname	Surname
1	Float away	The Springs	Pop	1	4.6.10	18:36	1	Sara	Bibi
2	In denial	Rocketts	Rock	2	5.6.11	19:20	1	Sara	Bibi
1	Float away	The Springs	Pop	3	9.6.12	09:30	2	Danny	Judd
2	In denial	Rocketts	Rock	4	1.7.14	08:28	2	Danny	Judd
1	Float away	The spings	Pop	5	9.1.11	03:45	3	Cara	Lichfield

Here is a **flat file database** that has been used to record data about the **downloads** of tracks from a music website.

An inefficient flat file database

TrackID	Title	Artist	Genre	DloadID	Date	Time	MemberID	Firstname	Surname
1	Float away	The Springs	Pop	1	4.6.10	18:36	1	Sara	Bibi
2	In denial	Rocketts	Rock	2	5.6.11	19:20	1	Sara	Bibi
1	Float away	The Springs	Pop	3	9.6.12	09:30	2	Danny	Judd
2	In denial	Rocketts	Rock	4	1.7.14	08:28	2	Danny	Judd
1	Float away	The spings	Pop	5	9.1.11	03:45	3	Cara	Lichfield

Activity

Use the **Activity 2 worksheet** to highlight any potential issues that might arise from storing the data in this way.

An inefficient flat file database

TrackID	Title	Artist	Genre	DloadID	Date	Time	MemberID	Firstname	Surname
1	Float away	The Springs	Pop	1	4.6.10	18:36	1	Sara	Bibi
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1	Float away	The Springs	Pop	3	9.6.12	09:30	2	Danny	Judd
2	In denial	Rocketts	Rock	4	1.7.14	08:28	2	Danny	Judd
1	Float away	The spings	Pop	5	9.1.11	03:45	3	Cara	Lichfield

Sara Bibi has downloaded two tracks. In a **flat file database**, her details would need to be entered **twice**. This is known as data **redundancy**.

An inefficient flat file database

TrackID	Title	Artist	Genre	DloadID	Date	Time	MemberID	Firstname	Surname
1	Float away	The Springs	Pop	1	4.6.10	18:36	1	Sara	Bibi
2	In denial	Rocketts	Rock	2	5.6.11	19:20	1	Sara	Bibi
1	Float away	The Springs	Pop	3	9.6.12	09:30	2	Danny	Judd
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1	Float away	The spings	Pop	5	9.1.11	03:45	3	Cara	Lichfield

The track '**Float away**' has been downloaded **three times**. This means that all of this data would need to be entered three times. Another example of **redundancy**.

An inefficient flat file database

TrackID	Title	Artist	Genre	DloadID	Date	Time	MemberID	Firstname	Surname
1	Float away	The Springs	Pop	1	4.6.10	18:36	1	Sara	Bibi
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1	Float away	The spings	Pop	5	9.1.11	03:45	3	Cara	Lichfield

Having to enter data **multiple times** means the database is at an increased risk of having inaccurate data.

An inefficient flat file database

TrackID	Title	Artist	Genre	DloadID	Date	Time	MemberID	Firstname	Surname
1	Float away	The Springs	Pop	1	4.6.10	18:36	1	Sara	Bibi
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It also makes it much harder to **update** because you would need to change **every occurrence** of the data item. This can lead to **inconsistent** data.

Relational databases

The most common model for a database is a **relational** model.

A **relational database** contains **more than one** table.

The data in the tables are linked using **relationships**.

Relational databases

Relationships can be:

- One to one
- One to many
- Many to many

Relational databases

A **one to one** relationship means that **one record** in a table relates to a **single** record in another table.

Relational databases

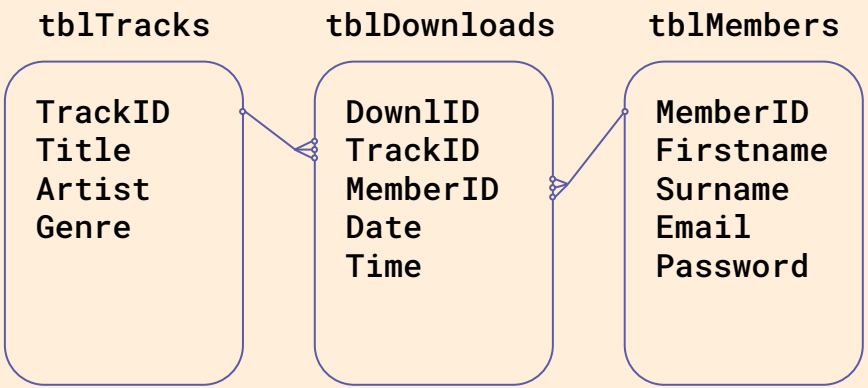
A **one to many** relationship means that **one record** in a table relates to **multiple** records in another table.

Relational databases

A **many to many** relationship means that **multiple** records in a table relate to **multiple** records in another table.

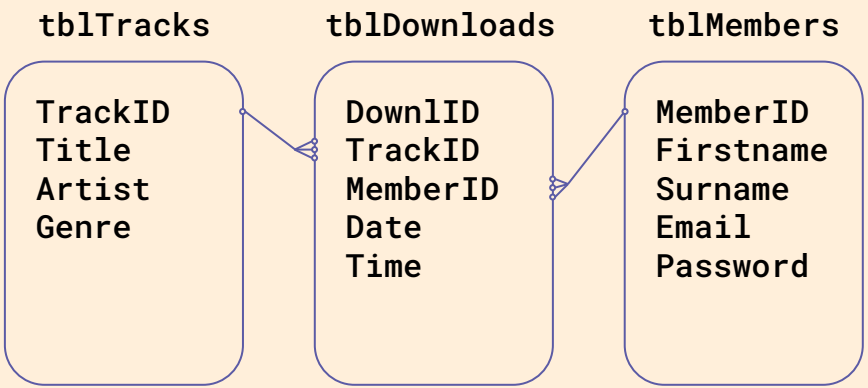
Relational databases

The downloads database needs to be stored in a **relational database**.



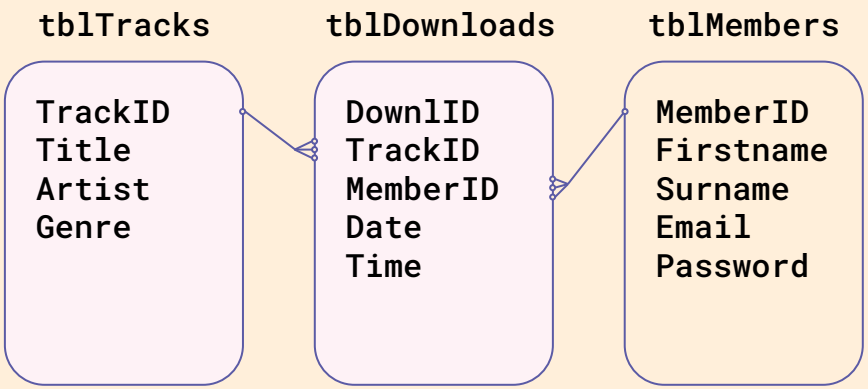
Relational databases

Keeping the database as a flat file database would cause issues with **inconsistency** and **data redundancy**.



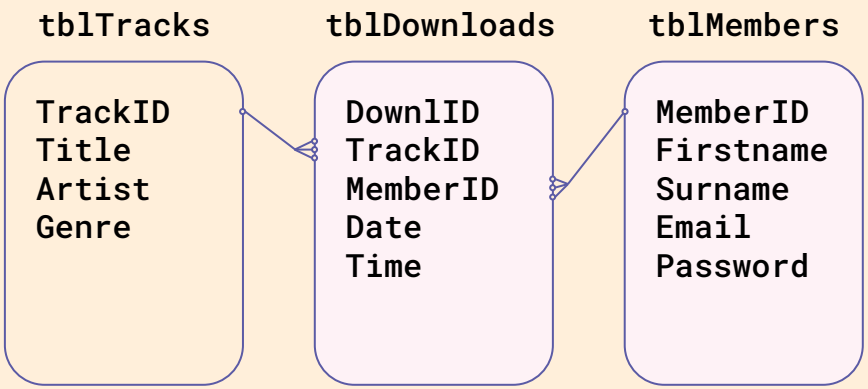
Relational databases

This is because each **track** can be **downloaded** multiple times.



Relational databases

And each **member** can have have multiple **downloads** of different tracks.

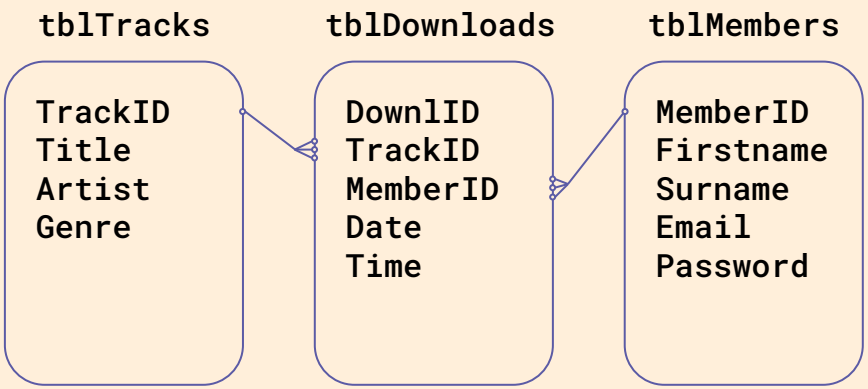


Relational databases

Question

What **relationships** are being formed here?

A. One to one
B. One to many
C. Many to many

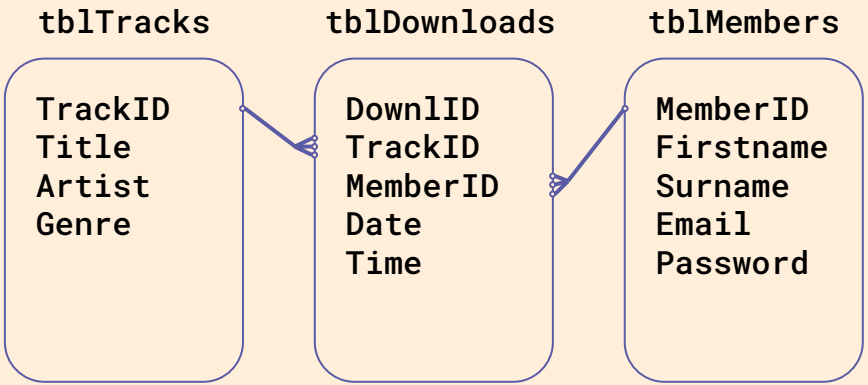


Relational databases

Question

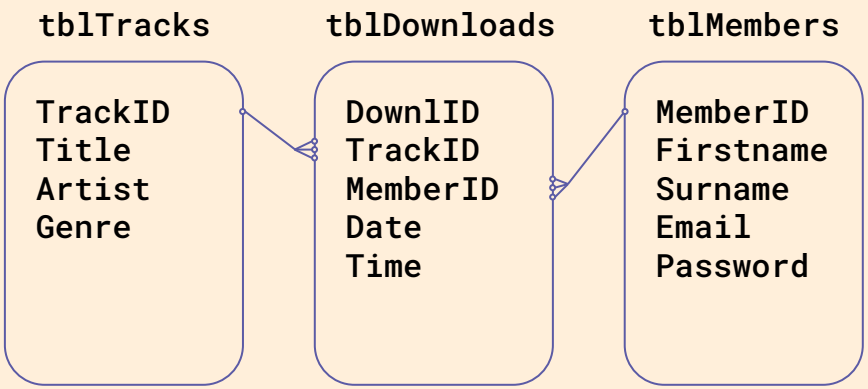
What **relationships** are being formed here?

A. One to one
B. **One to many**
C. Many to many



Relational databases

The other benefit of a **relational database** is that all data can be updated at the original **source**.



Relational databases

Data doesn't need to be repeated and relationships can be made using the source table's **unique identifier**.

tblDownloads

DownlID	TrackID	MemberID	Date	Time
1	1	1	1.5.18	03:40
2	2	1	2.9.20	08:29
3	2	2	3.4.20	11:25
4	3	2	1.8.19	13:45

Relational databases

A **primary key** is unique to the original source of the data.

It **cannot** be repeated.

tblDownloads

DownlID	TrackID	MemberID	Date	Time
1	1	1	1.5.18	03:40
2	2	1	2.9.20	08:29
3	2	2	3.4.20	11:25
4	3	2	1.8.19	13:45

Relational databases

When you link to a source table's primary key you use a **foreign key**.

tblDownloads

DownlID	TrackID	MemberID	Date	Time
1	1	1	1.5.18	03:40
2	2	1	2.9.20	08:29
3	2	2	3.4.20	11:25
4	3	2	1.8.19	13:45

Relational databases

A **foreign key** *can* be repeated because it is a link back to the **primary key** in the source table.

tblDownloads

DownlID	TrackID	MemberID	Date	Time
1	1	1	1.5.18	03:40
2	2	1	2.9.20	08:29
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Relational databases

Using the **foreign key** you can go to the **tblTracks** table and find the linked track.

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Relational databases

You can also use the **foreign key** for the **member** to find out who downloaded that track at that time on that day.

tblDownloads

DownlID	TrackID	MemberID	Date	Time
1	1	1	1.5.18	03:40
2	2	1	2.9.20	08:29
3	2	2	3.4.20	11:25
4	3	2	1.8.19	13:45

Relational databases

You can also use the **foreign key** for the **member** to find out who downloaded that track at that time on that day.

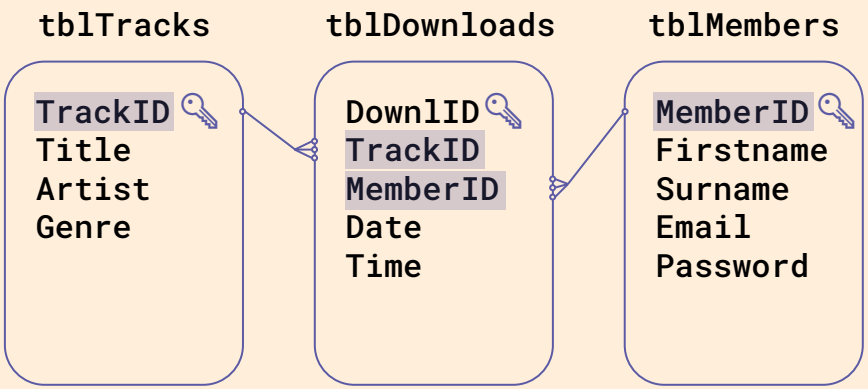
tblMembers

MemberID	Firstname	Surname	Email	Password
1	Sara	Bibi	s@b.com	123456
2	Danny	Judd	d@j.com	987654
3	Cara	Lichfield	c@l.com	qwerty

Relational databases

The music database now has **three tables**. All linked together using **relationships**.

Data isn't repeated because it is linked to the source table's **primary key** using the **foreign key**.



What other data has been repeated?

Take another look at the music database.

Questions Think, write, pair, share

Where can you still see repetition of data?

What could you do to reduce the repetition?

What other data has been repeated?

Take another look at the music database.

Answers Think, write, pair, share

Where can you still see repetition of data?

Genre and Artist are repeated in the tracks table.

What other data has been repeated?

Take another look at the music database.

Answers Think, write, pair, share

What could you do to reduce the repetition?

You could have separate tables for genres and artists.

In this lesson, we...

Learnt about flat file and relational databases.

Learnt the key terms associated with databases.